

unexecuted shares. Cash balancing for risk-management was originally implemented when investors did not have full confidence in the underlying intraday covariance model.

Objective Function Difficulty

The formulation of the objective function above presents many difficulties. First, the problem is not linear or quadratic, thus creating increased complexity for the optimization routine. Second, the timing risk component is represented as a square root function as opposed to a squared term in a quadratic programming (QP) optimization problem. For example, portfolio construction optimization models often express risk as a variance term (risk squared) and can be directly incorporated into quadratic optimizations. Finally, there are $n \times m$ decision variables in our full formulation—one decision variable for each stock in each trading period. In portfolio construction there are m decision variables—one for each stock. Our portfolio optimization requires m times more solution variables.

Unfortunately, the time to solve these optimization algorithms increases at an exponential rate with the number of variables. For a 500-stock portfolio executed over 26 trading intervals (e.g., 15 minute intervals) this results in 13,000 decision variables and takes much more than 26 times longer to solve. Combined with the constraints above, it makes solving this optimization extremely slow.

Investors require accurate solutions within a short enough timeframe to be useful for trading. By “short enough” we mean a matter of seconds or minutes as opposed to minutes or hours.

Fortunately, there are accurate transformations and approximations that allow us to solve the trader’s dilemma in a reasonable amount of time. These techniques are:

- Quadratic Optimization Approach
- Trade Schedule Exponential
- Residual Schedule Exponential
- Trade Rate Optimization

Optimization Objective Function Simplification

The full portfolio optimization objective function ([Equation 9.5](#)) includes permanent market impact cost. Because this cost is not dependent upon the specified trading strategy we can omit it from the objective function without changing the optimal solution. However, permanent impact needs